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Tumors of the Urethra

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Keywords

inguinal lymph nodes; radical cystectomy; squamous cell carcinoma; urethral cancer; urethrectomy

Questions

1. Which of the following is not a possible origin of female urethral adenocarcinoma?
    a. Müllerian tissue
    b. Cowper glands
    c. Glandular metaplasia
    d. Skene glands
    e. Urothelium
2. The most common histologic subtype of male anterior urethral cancer is:
    a. adenocarcinoma.
    b. squamous cell carcinoma.
    c. urothelial carcinoma.
    d. melanoma.
    e. sarcoma.
3. The histologic subtype of urethral cancer that is more common in women than in men is:
    a. adenocarcinoma.

b. squamous cell carcinoma.
c. urothelial carcinoma.
d. melanoma.
e. sarcoma.

4. In male pendulous urethral carcinoma:
   a. prognosis depends on histologic cell type.
   b. surgical excision alone is often curative.
   c. prognosis is worse than for bulbomembranous urethral cancer.
   d. a 2-cm margin is always required for penile preserving surgery.
   e. biopsy most commonly demonstrates transitional cell carcinoma.
5. A treatment that is NOT indicated in cT3N0M0 squamous cell carcinoma of the bulbar urethra is:
   a. radical cystectomy with urethrectomy.
   b. chemoradiotherapy.
   c. inguinal lymphadenectomy.
   d. pelvic MRI.
   e. neoadjuvant chemotherapy.
6. The strongest risk factor for male urethral recurrence after radical cystectomy is:
   a. carcinoma in situ.
   b. bladder tumor multifocality.
   c. prostatic urethral involvement.
   d. squamous cell histology.
   e. prior nonmuscle invasive bladder cancer.
7. Which of the following is FALSE about urethral recurrence after radical cystectomy and orthotopic neobladder?
   a. Urethral recurrences are more common after radical cystectomy and ileal conduit.
   b. Radical urethrectomy is always required.
   c. Transurethral resection and intraurethral bacille Calmette-Guérin (BCG) can be effective for noninvasive tumors.
   d. Bowel from the neobladder can be reconfigured for a cutaneous diversion.
   e. Patients with noninvasive recurrences have a good prognosis.

8. A 71-year-old man is diagnosed with a primary prostatic urothelial carcinoma. Transurethral biopsy demonstrates invasion into the prostatic stroma, and imaging is negative for metastasis. The best treatment option is:
    a. systemic immunotherapy.
    b. aggressive transurethral resection of the prostate followed by BCG.
    c. radical cystectomy and urethrectomy with neoadjuvant chemotherapy.
    d. whole pelvic radiation.
    e. radical prostatectomy.
9. When a delayed urethrectomy is performed after radical cystectomy in a male, the most important technical consideration to minimize the risk of a local recurrence is:
    a. removal of the entire distal urethra including the meatus.
    b. bilateral groin dissections.
    c. total penectomy.
    d. intraoperative ultrasound.
    e. neoadjuvant radiation.
10. A condition that does not increase the risk of female urethral carcinoma is:
    a. leukoplakia.
    b. bacterial vaginosis.
    c. urethral diverticulum.
    d. urethral stricture disease.
    e. human papillomavirus infection.
11. The strongest prognostic factor for survival in female urethral cancer is:
    a. tumor stage.
    b. age at presentation.
    c. histologic subtype.
    d. multimodal treatment.
    e. urinary retention.
12. A 68-year-old man presents with an anterior urethral tumor invasive into the corpora cavernosum. His tumor stage is:
    a. T1.

b. T2a.
c. T2b.
d. T3.
e. T4.

13. The best treatment for a small, exophytic cT2N0 female urethral carcinoma located near the urethral meatus is:
    a. excision of the distal 2/3 of the urethra.
    b. transvaginal biopsy followed by systemic chemotherapy.
    c. radical urethrectomy with bladder neck closure and ileovesicostomy.
    d. chemoradiation.
    e. inguinal lymphadenectomy.
14. A 76-year-old female presents with a locally invasive proximal urethral tumor abutting the pubic symphysis. The treatment that would be inappropriate is:
    a. radical cystectomy with pubectomy.
    b. external beam radiation with systemic chemotherapy.
    c. radical cystectomy with intraoperative radiation.
    d. radiation monotherapy.
    e. chemoradiation followed by surgical consolidation.
15. Two years after a radical cystectomy and ileal conduit in a male patient, a urethral wash cytology is positive for high-grade urothelial carcinoma. The next step is:
    a. repeat urethral wash cytology.
    b. urethroscopy with biopsy.
    c. radical urethrectomy.
    d. pelvic MRI.
    e. urine cytology.
16. All of the following are true about the male urethra EXCEPT:
    a. the anterior urethra drains into the inguinal lymph nodes.
    b. the glands of Littre line the anterior urethra.
    c. the posterior urethra consists of the prostatic and membranous urethra.
    d. the pseudostratified epithelium of the pendulous urethra transitions to glandular epithelium in the fossa navicularis.

e. anterior urethral cancers are more common in the bulbar urethra.

17. A 56-year-old-man presents with an invasive pendulous urethral squamous cell carcinoma with palpable inguinal lymph nodes bilaterally. Imaging is negative for distant metastasis. The best treatment is:
    a. radiation monotherapy to the tumor and lymph nodes.
    b. chemoradiation to the penis and groins followed by surgical consolidation.
    c. radical penectomy.
    d. dynamic sentinel lymph node biopsy.
    e. distal urethrectomy with 6 weeks of oral antibiotics.

Answers

1. **b. Cowper glands.** There are several proposed origins of female urethral adenocarcinoma, and more than one tissue may be involved. All of the listed tissue types are possible origins with the exception of Cowper glands, which are only present in men.
2. **b. Squamous cell carcinoma.** The male anterior urethra is lined by stratified and pseudostratified columnar epithelium, which transitions to squamous epithelium distally. The majority of male anterior urethral carcinomas are squamous cell carcinoma.
3. **a. Adenocarcinoma.** Gender is not independently associated with survival for urethral carcinoma; however, there are differences in the characteristics of male and female urethral carcinoma, including the distribution of histologic subtypes. Although squamous cell carcinoma is the predominant form in men, women are more likely to have adenocarcinoma compared to men.
4. **b. Surgical excision alone is often curative.** The prognosis of male pendulous anterior urethral cancer is strongly associated with stage, is more favorable than bulbomembranous urethral cancer, and is unrelated to histologic cell type. Most pendulous urethral tumors are squamous cell carcinomas. Low-stage pendulous anterior urethral cancers can often be cured with surgical excision alone,

and penile preservation can often be achieved with margins as little as 5 mm.

5. **c. Inguinal lymphadenectomy. Locally advanced bulbomembranous urethral carcinoma requires aggressive treatment.** MRI is required to fully stage the disease, including the extent of local invasion and presence of pelvic nodal metastasis. Of the treatments listed, neoadjuvant chemotherapy, chemoradiation, and radical cystectomy with urethrectomy are all reasonable options. The role of inguinal lymphadenectomy for cN0 urethral cancer is controversial, but there is no role for this in bulbomembranous disease as these tumors are unlikely to drain into the inguinal lymph nodes.
6. **c. Prostatic urethral involvement.** Urethral recurrence after radical cystectomy occurs in 5% to 10% of patients. Of the multiple risk factors that have been studied, prostatic urethral involvement, a positive urethral margin, and cutaneous urinary diversions have the highest risk of urethral recurrence.
7. **b. Radical urethrectomy is always required.** Patients with large or invasive urethral recurrences after radical cystectomy require aggressive treatment, usually with radical urethrectomy. If a urethral recurrence occurs after an orthotopic neobladder, the bowel from the neobladder can be used for a cutaneous diversion in many cases. Patients with noninvasive urethral recurrences have a favorable prognosis and can often be managed with transurethral resection and topical BCG, thus avoiding radical urethrectomy in some cases. Prostatic urethral involvement, a positive urethral margin, and cutaneous diversions all increase the risk of urethral recurrences.
8. **c. Radical cystectomy and urethrectomy with neoadjuvant chemotherapy.** The prognosis of primary prostatic urethral carcinoma is strongly associated with depth of invasion into the prostate. Those with superficial or lamina propria invasion can often be managed with transurethral resection and BCG; however, those with tumors invasive into the stroma require aggressive treatment. Radical cystoprostatectomy with urethrectomy and preoperative chemotherapy is the best choice.

9. **a. Removal of the entire distal urethra including the meatus.** A delayed urethrectomy after radical cystectomy can be a challenging procedure. Critically, the entire urethra including the fossa navicularis and urethral meatus should be removed en bloc to eliminate the risk of a distal recurrence, even for proximal tumors. There is no role for routine inguinal lymphadenectomy, and penectomy is not required unless the urethral tumor is invasive into the deep structures of the penis.
10. **b. Bacterial vaginosis.** Urethral carcinoma is most likely caused by conditions that cause inflammation of the urethra, such as leukoplakia, urethral strictures, and HPV infection. Female urethral diverticula can increase the risk of urethral cancers, most commonly adenocarcinomas. Bacterial vaginosis has no known association with urethral cancer.
11. **a. Tumor stage.** The prognosis of female urethral cancer is strongly associated with stage at diagnosis. Tumors that arise from the proximal urethra tend to present at a more advanced stage. There is no known association between prognosis and histologic subtype, age at presentation, or presenting symptoms.
12. **d. T3.** Male anterior urethral cancer with invasion into the corpora cavernosum or anterior vagina are classified as T3 by the AJCC 8th Edition Staging System. Tumors invasive into the corpus spongiosum or periurethral tissue are classified as T2. The AJCC staging system does not have a T2a or T2b for urethral cancer.
13. **d. Chemoradiation.** Small exophytic distal female urethral cancers can be effectively treated with surgical excision or radiation with or without chemotherapy. Radical urethrectomy can usually be avoided for distal tumors. The distal 1/3 of the female urethra can be safely excised without compromising urinary continence, but excision of the distal 2/3 will likely cause incontinence and is not a good option. There is currently no role for prophylactic inguinal lymphadenectomy for cN0 female distal urethral cancer. Most distal urethral tumors can be biopsied transurethrally.
14. **d. Radiation monotherapy.** Proximal female urethral cancer often presents at an advanced stage, tends to be aggressive, and can have a poor prognosis. Therefore, aggressive local therapy is required,

often in a multidisciplinary setting. Of the listed options, radiation monotherapy is inadequate treatment.

15. **b. Urethroscopy with biopsy.** Urethral wash cytology may be beneficial for detecting early urethral recurrences after radical cystectomy. However, a positive cytology can be falsely positive and always requires confirmation with a biopsy prior to aggressive treatment or further imaging.
16. **d. The pseudostratified epithelium of the pendulous urethra transitions to glandular epithelium in the fossa navicularis.** All of the listed options are true except "d," as the urethral epithelium transitions to squamous, not glandular, in the fossa navicularis.
17. **b. Chemoradiation to the penis and groins followed by surgical consolidation.** Advanced staged urethral carcinoma requires aggressive multimodal treatment. Clinically positive inguinal lymph nodes with an invasive primary urethral cancer should be considered metastatic disease, therefore there is no role for antibiotics or sentinel lymph node biopsy. Radiation monotherapy and surgical monotherapy to the penis alone are both suboptimal treatment for invasive, cN+ penile cancers. Chemoradiation to the penis and groins with surgical consolidation is the best choice.

Chapter Review

1. The prognosis of distal anterior male posterior urethral cancer (PUC) is better than that of proximal anterior PUC, which is often associated with local invasion and distant metastasis.
2. Multimodal therapy should be considered for advanced urethral tumors given the poor outcomes after surgery and radiation alone.
3. The benefit from prophylactic inguinal lymph node dissection has not been demonstrated in anterior urethral cancer.
4. Primary posterior urethral carcinoma is rare, and noninvasive tumors can be effectively managed with transurethral resection of the prostate and BCG.
5. The three most common histologies for female urethral cancer are urothelial carcinoma, squamous cell carcinoma, and

adenocarcinoma.

6. Compared to anterior urethral cancers, posterior cancers present at a more advanced stage and are associated with worse survival.
7. Radiation and surgical excision are both options for low-stage anterior tumors, each with high cure rates.
8. Surgery and radiation therapy for proximal female urethral tumors have poor outcomes when used alone, therefore multimodal therapy is recommended.
9. Male urethral recurrence after radical cystectomy is rare, but more common with prostatic involvement, a positive urethral margin, and cutaneous diversions.
10. The primary risk factors for female urethral recurrences are bladder neck involvement and a positive urethral margin.
11. Most surgeons will proceed with orthotopic neobladder if an intraoperative frozen section of the urethral margin is negative.
12. After urethrectomy in patients with orthotopic neobladders, bowel from the neobladder can often be reconfigured and used for a cutaneous diversion.

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Inguinal Node Dissection

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Keywords

inguinal lymph node dissection; lymph node metastases; papilloma virus; penile cancer; penile tumor; squamous cell carcinoma

Questions

1. For lymphatic spread in penile cancer, which of the following statements is true?
    a. Penile carcinoma spreads initially to inguinal lymph nodes before the development of distant metastatic disease.
    b. There is significant crossover in the lymphatic drainage of the penis; hence bilateral inguinal lymph node dissection (ILND) is usually recommended.
    c. Lymph node metastases confers a poor prognosis overall when present for patients with penile carcinoma.
    d. Lymphovascular invasion or greater than 50% poorly differentiated tumors are at a higher risk for lymph node metastases.
    e. All of the above.
2. Preservation of the fatty layer within Camper fascia is directly associated with:
    a. infection rates after the surgery.
    b. lymphatic complications due to preservation of the lymphatics.

c. blood supply of the overlying skin.
d. preservation of the Innervation of the overlying skin.
e. all of the above.

3. Which of the following imaging techniques are mandatory in the presurgery assessment of a patient with penile carcinoma for the inguinal region?
    a. Ultrasound with fine needle aspiration cytology (FNAC)
    b. Computed tomography (CT)
    c. Magnetic resonance imaging (MRI)
    d. MRI with nanotechnology
    e. Positron emission tomography (PET scan)
    f. None of the above
4. Which of the following statements are considered to be true for ILND?
    a. It is not useful because it overtreats around 80% of the patients that are at risk of surgical complications.
    b. It should be performed only when lymphatic affectation is clinically evidenced to avoid further spread of the disease.
    c. Has been substituted by the Sentinel Node biopsy proposed by Cabanas et al. because it decreases the complication rate, adding a good diagnostic assessment.
    d. It's useful for penile staging/treatment because 17% to 25% of patients with nonpalpable lymph nodes will have lymphatic metastases, but carries a high morbidity burden.
5. Which of the following statements is true regarding dynamic sentinel node biopsy (DSNB)?
    a. After a positive DSNB, patients should undergo strict active vigilance with PET scans and repeated biopsies every 6 months.
    b. DSNB has a relatively short learning curve that helped this procedure to be widely adopted in several centers around the world, avoiding unnecessary ILND.
    c. When properly done, DSNB can achieve more than 80% of sensitivity for lymph node metastases in patients with penile cancer.

d. On the DSNB, there is only the use of technetium nanocolloid the day before the procedure described.
e. All of the above.

6. Modifications of the original technique/template for inguinal lymph node dissection has been proposed, although no randomized clinical trials have been performed to compare them. Which of the following statements is the rationale for superficial inguinal node dissection and modified inguinal lymph node dissection (MILD)?
    a. All nodes affected in patients with penile cancer will be located within the reduced template whilst reducing morbidity.
    b. Frozen section is reliable for detection of lymph nodes affected.
    c. Preserving the saphenous vein decreases the complication rate.
    d. If nodes in the superficial compartment are negative, there should not be affectation of the deep package.
7. Based on the information gathered by previous series, what can be stated from the endoscopic inguinal lymph node dissection?
    a. It has to be performed in a two-staged fashion to perform both legs.
    b. It has been shown to have the same complication rate in most of the open series.
    c. Oncologically, it seems to be equivalent to open procedures.
    d. The limits of the triangle are the inguinal ligament, the abductor longus, and the fascia of the iliacus muscle.
    e. All of the above.
8. Which of the following statements is true regarding the robotic and laparoscopic ILND?
    a. Robotics have been proved to have less complications due to a more precise dissection.
    b. The 3D visualization that the robotic system offers allow the surgeon to identify more lymphatic nodes.
    c. The degrees of movements that the robotic system instruments have allows the surgeon to have easier control of the lymphatics with ultrasonic energy.
    d. The Xi platform allows repositioning without modifications from the operative room setting.

e. None of the above.

9. During a radical ILND, sartorius muscle interposition is carried out in order to:
    a. improve the mobility of the lower limb after surgery.
    b. be used as an interposition tissue to avoid the formation of collections.
    c. protect the femoral vessels.
    d. none of the above.
    e. all of the above.
10. Which type of complication decreased in frequency as a result of the adoption of endoscopic/robotic techniques for ILND in penile cancer?
    a. Wound complications
    b. Lymphatic complications
    c. Bleeding complications
    d. Major complications that resulted in the death of the patient
    e. Deep venous thrombosis
11. What is the 5-year survival rate of a patient who, after ILND, was found to have 1 to 3 positive lymph nodes for malignancy?
    a. 5% to 15%
    b. 75% to 80%
    c. 90% to 95%
    d. 60% to 70%
    e. 30% to 50%
12. What is the overall 5-year survival rate of a patient with positive pelvic lymph nodes for malignancy?
    a. Less than 10%
    b. 25%
    c. 35%
    d. 45%
    e. 55%

Answers

1. **e. All of the above. Knowledge on the lymphatic anatomy and spread of the penile tumors is crucial to understand the rationale for its surgical management.** In this question, some critical points are highlighted. First is the fact that penile cancer usually spreads in a stepwise manner going from the inguinal nodes (superficial package first, then deep package) and pelvic nodes before the appearance of the distant metastatic lesions. However, due to the crossover of the lymphatic channels in the symphyseal region, the ILND should be performed bilaterally independently of the side affected. Furthermore, positive lymph nodes have proved to be a prognostic survival factor in several studies, which underlines the importance of undertaking this procedure when indicated. One of the indications for ILND are high-risk tumors; these are defined as T1 with high-grade and/or lymphovascular invasion and/or >50% poorly undifferentiated. All T2 and T3 tumors are considered high risk.
2. **c. Blood supply of the overlying skin.** Irrigation of the skin flaps raised during the dissection depends on anastomotic vessels in the superficial fatty layer of Camper fascia. Because lymphatic drainage of the penis to the groin runs beneath Camper fascia, this layer can be preserved and left attached to the overlying skin when the working space is created.
3. **f. None of the above.** There are not imaging techniques formally recommended for the evaluation of the inguinal area in patients with penile cancer. Nonetheless, CT of abdomen, pelvis, and chest is recommended in patients with ILND involvement or in patients with clinically suspected pelvic lymph nodes.
4. **d. It's useful for penile staging/treatment because 17% to 25% of patients with nonpalpable lymph nodes will have lymphatic metastases, but carries a high morbidity burden.** ILND in patients with micrometastasis has shown to increase survival, 17% to 25% of the patients with clinically negative nodes will harbor micrometastasis. However, due to the morbidity burden that this represents for some patients, appropriate patient selection is crucial to minimize morbidity without sacrificing oncological outcomes. Early ILND has shown to be beneficial concerning patients'

survival, which is why it is not recommended to delay the surgery until lymphatic affectation is evidenced. Cabanas sentinel node has been disregarded due to lack of sensitivity in the diagnosis. However, dynamic sentinel node in expert hands could be an option for proper patient selection (see “Penile Cancer: Non-Palpable Inguinal Adenopathy”).

5. **c. When properly done, DSNB can achieve more than 80% of sensitivity for lymph node metastases in patients with penile cancer. After positive DSNB, patients should undergo ILND.** This modality has shown to be effective in diagnosing patients with occult metastases. However, it has a long learning curve, which is why its worldwide adoption have been troublesome. In expert hands, it has shown to have an 80% sensitivity rate. One of the factors that this technique includes is the use of the nanocolloid and the blue dye, to improve results.
6. **d. If nodes in the superficial compartment are negative, there should not be affectation of the deep package.** Both templates share the fact that if nodes are negative in the superficial package, the chances of having positive deep lymph nodes are scarce. However, there has been reports of positive nodes in the superficial space outside the MILND template. The intraoperative frozen section can miss more than 25% of positive nodes, and definitive proof of diminishment of the morbidity with the preservation of the saphenous veins has not been given; reports on this regard are contradictory.
7. c. **Oncologically, it seems to be equivalent to open procedures.** Endoscopic ILND has been shown to be oncologically safe and equivalent by showing no differences in the number of positive nodes when comparing endoscopic versus open dissections. Endoscopic ILND can be performed in one time in both legs; it has shown a decreased complication rate (no randomized clinical trial has been done on this regard), and the limits are the inguinal ligament, the abductor longus, and the sartorius muscle.
8. **d. The Xi platform allows repositioning without modifications from the operative room setting.** Robotics has not shown superiority over the laparoscopic approach (not due to to the

increased degrees of movements that allow for more precise dissection nor to the tridimensional view). However, Xi facilitates the repositioning of the robot when compared to other platforms such as the Si. This is possible in the Si system, but reaccommodation of the operative room is needed.

9. **c. Protect the femoral vessels.** Sartorius muscle had been used with the intention of protecting the femoral vessels. However, it has shown to be a predictor factor for complications in some series.
10. **a. Wound complications.** With a reduced incision size, decreased risk for skin/wound complications is posed on the patients when an endoscopic approach is used. This is the rationale for the endoscopic approach. Lymphatic complications are more related to the type of energy used in the lymph node management strategy used along with the postoperative care. Same for bleeding complications. Other complications should not have a difference between approaches. Although, no randomized clinical trial has been done on this regard.
11. **b. 75% to 80%. Seventy-five percent to eighty percent is the 5-year survival rate of a patient who, after ILND, was found to have one to three positive lymph nodes for malignancy, which highlights the importance of a prompt ILND and the number of lymph nodes involved in patients with penile cancer.**
12. **a. Less than 10%.** Less than 10% of patients with positive lymph pelvic nodes, bilaterally positive nodes, and/or extranodal extension survive after 5 years.

Chapter Review

1. The most important independent prognostic factor related to cancer-specific survival in squamous cell carcinoma (SCC) of the penis is lymph node metastasis.
2. Immediate ILND is associated with improved survival compared to salvage surgery for clinically detectable disease.
3. For centers of excellence with expertise in penile cancer and ILND, DSNB can be performed with a false-negative rate of less than 5%.

4. Superficial and modified ILND are options in the initial management of inguinal region in patients with nonpalpable disease. These are techniques have less morbidity than radical ILND.
5. Endoscopic inguinal lymphadenectomy has been demonstrated to have less morbidity than open ILND and to remove comparable number of lymph nodes.
6. Radical ILND is indicated for palpable disease, with potential for cure in inguinal-only disease, and for palliation of pain, infection, or hemorrhage in selected patients.
7. Adjuvant chemotherapy can aid in better outcomes in patients with positive inguinal disease.
8. Pelvic metastasis do not occur in patients with negative inguinal nodes.

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Surgery for Benign Disorders of the Penis and Urethra

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Keywords

flaps; genital reconstruction; genital trauma; grafts; penile amputation; penile reconstruction; phalloplasty; urethral reconstruction

Questions

1. In terms of tissue transfer, which of the following statements concerning grafts is TRUE?
    a. The process of take is less than 48 hours.
    b. A graft is tissue that is excised from a donor site and reestablishes its blood supply by revascularization.
    c. During imbibition, the first phase of take, the graft exists above body temperature.
    d. Conditions of take are a reflection of only the graft host bed.
    e. Split-thickness skin grafts (STSGs) are less prone to contraction compared with full-thickness grafts.
2. With regard to the microanatomy of grafts, using skin as a model, which of the following statements is TRUE?
    a. The intradermal plexus is at the interface of the superficial and deep dermis.

b. The subdermal plexus is carried at the juncture of the deep dermis and the underlying tissue.
c. The lymphatics are most richly distributed in the adventitial dermis.
d. The adventitial dermis, because of its collagen content, accounts for the majority of the physical characteristics.
e. Genital and extragenital skin behave similarly when used in genitourinary reconstruction.

3. With regard to the grafts used most commonly in genitourinary reconstructive surgery, which of the following statements is TRUE?
   a. Full-thickness skin is an optimal replacement for the tunica albuginea of the corpora cavernosa.
   b. Bladder epithelial graft is fastidious because of the nature of the superficial lamina.
   c. Buccal mucosa graft is thought to have a panlaminar plexus.
   d. Tunica vaginalis graft has proved to be a very reliable one for single-stage urethral reconstruction.
   e. Buccal mucosa grafts should not be thinned because it will adversely affect the physical and vascular properties of the graft.
4. With regard to the anatomy of the penile shaft, which of the following statements is TRUE?
   a. Throughout most of the length of the penis, the septum is a true competent septum.
   b. The erectile tissues of the normal corpora cavernosa are separated from the tunica by the space of Smith.
   c. The dorsal arteries of the penis are carried in an envelope fashion in the dartos fascia.
   d. The Buck fascia is loosely areolar and lies immediately beneath the skin.
   e. The Buck fascia attenuates on the ventrum, lateral to the corpus spongiosum.
5. According to consensus, the urethra should be divided into six entities. Which of the following statements is most accurate?
   a. The fossa navicularis is that portion of the urethra that is most dorsally displaced with regard to the surrounding spongy

erectile tissue.

b. The bulbous urethral portion is invested by the thickest portion of the corpora spongiosum.
c. The bulbous urethral at its proximal extent is part of the posterior urethra.
d. The membranous urethra is invested by the most proximal aspect of the corpus spongiosum.
e. The membranous urethra, throughout its length, is surrounded by the external rhabdosphincter.

6. With regard to the arterial vascularization of the deep structures of the penis, which of the following statements is TRUE?
   a. The circumflex cavernosal arteries are uniform in number and distribution.
   b. The arteries to the bulb arborize into the spongy erectile tissue of the glans.
   c. The common penile artery represents the end continuation of the superficial external pudendal artery.
   d. The common penile artery divides to become the cavernosal artery and the dorsal arteries, after branching off the circumflex cavernosal arteries.
   e. The blood supply of the deep structures of the penis is derived from the common penile artery, which is the continuation of the internal pudendal artery after it branches off its perineal branch and the posterior scrotal arteries.
7. With regard to the innervation to the penis, which of the following statements is TRUE?
   a. The cavernosal nerves are purely parasympathetic and are the extensions of the nervi erigentes.
   b. The pudendal nerves accompany the dorsal artery of the penis and the dorsal vein of the penis as they run through the obturator foramen.
   c. The dorsal nerve arises in the Alcock canal as a branch of the pudendal nerve.
   d. The dorsal nerves throughout their course are prominent, large nerve bundles.

e. The skin of the shaft of the penis is innervated by a branch of the femoral nerve.

8. Which of the following statements regarding lichen sclerosus (LS) is FALSE?
    a. Lichen sclerosus is the preferred term for what was previously known as *balanitis xerotica obliterans.*
    b. LS is the most common cause of meatal stenosis.
    c. There is a strong association with an infectious etiology for the development of LS.
    d. Circumcision may be curative if only foreskin is involved with LS.
    e. Topical steroids may help stabilize the inflammatory process early in the disease course.
9. Each of the following statements regarding complications associated with LS are true, EXCEPT:
    a. Urethral stricture may develop secondary to iatrogenic injury from repeat instrumentation.
    b. Genital or extragenital skin may be used for urethral reconstruction for LS-associated urethral strictures.
    c. Meatal stenosis and high-pressure voiding with subsequent urinary extravasation into the glands of Littre may result in urethral stricture formation.
    d. Squamous cell carcinoma may develop in patients with a long history of LS.
    e. Patients under consideration for meatal reconstruction with LS must undergo retrograde urethrography to rule out proximal strictures before surgery.
10. Urethrocutaneous fistulae may develop secondary to all of the following, EXCEPT:
    a. Complication following hypospadias repair or other urethral surgery
    b. Recurrence following fistula repair as a result of distal obstruction and high-pressure voiding
    c. Extravasation of infected urine and periurethral abscess formation
    d. Complication of genital herpes simplex viral infection

e. Early fistula following urethral surgery may result from hematoma, infection, or tension with the closure

11. With regard to urethral meatal stenosis in childhood, which of the following statements is TRUE?
    a. Meatal stenosis is a frequent complication of phimosis.
    b. Meatal stenosis is frequently associated with upper tract changes, and all patients should be evaluated with ultrasonography and voiding cystourethrography.
    c. When ammoniacal meatitis is noted, often a short course of meatal dilation and steroid cream application will resolve the problem.
    d. When meatal stenosis is present, usually a dorsally based YV advancement flap repair is preferred.
    e. Meatal stenosis in childhood is frequently associated with concomitant LS.
12. When treating a patient with penile amputation, which of the following statements is TRUE?
    a. Replantation is not a consideration in self-inflicted injury, because most of these patients are chronically psychotic and will eventually try to amputate the penis again.
    b. If the distal part of the penis is not available, even if the amputation involves mostly skin with much of the shaft preserved, it is recommended that the remaining shaft be buried in the scrotum.
    c. In the case of amputation associated with avulsion, debridement to undivided tissue must precede penile replantation.
    d. The classic technique for replantation involves coaptation of the dorsal nerve, the deep dorsal vein, and the cavernosal arteries.
    e. The McRoberts technique of macroreplantation is not the preferred method of management for these patients, but when the situation warrants it, it is very successful.
13. All of the following statements regarding circumcision are true, EXCEPT:

a. Circumcision may be performed in the neonatal period in newborns born with a distal hypospadias, but not in those with proximal hypospadias.
b. Young boys with recurrent urinary tract infections (UTIs) should be considered for circumcision.
c. The most common complication following newborn circumcision is bleeding.
d. Circumcision has been shown to reduce the risk of human immunodeficiency virus (HIV) transmission in heterosexual men.
e. Small skin dehiscence following circumcision may be managed conservatively with local wound care and healing by secondary intention.

14. Concerning genital lymphedema, which of the following statements is TRUE?
    a. Reconstruction for lymphedema that is the consequence of the indirect effects of radiation is best accomplished with excision of the tissues and coverage with STSGs.
    b. In reconstruction for lymphedema, it is essential to maintain the parietal tunica vaginalis of the testes intact with grafting over that location.
    c. When considering reconstruction for lymphedema, full-thickness skin grafts (FTSGs) are preferable because of the distribution of the lymphatics in the superficial (adventitial) dermis.
    d. In the case of genital lymphedema, it is not unusual for the immune response of the tissues to be altered and for patients to have significant involvement with genital papillomas.
    e. Not unusually, in cases where the genital lymphedema is localized, the midline of the scrotum can be preserved for reconstruction.
15. Which of the following statements is MOST accurate concerning anterior urethral stricture disease?
    a. It causes limitation of the urethral lumen because of the bulk of the scar.
    b. It most often is limited to the urethral epithelium.

c. It implies a scarring process, usually involving both the epithelium and the underlying spongy erectile tissue of the corpora cavernosa.
d. It causes limitation of the urethral lumen because of contraction and noncompliance of the scar.
e. It is a metaplastic process of the urethral epithelium.

16. Which of the following statements concerning pelvic fracture urethral injury (PFUI) is TRUE?
    a. It involves the tissues of the epithelium as well as the underlying erectile tissues of the corpora cavernosa.
    b. It involves the epithelium, as well as the underlying erectile tissue of the corpus spongiosum.
    c. It is not a true stricture but rather fibrosis that results from distraction of the urethra.
    d. The stricture process can often be occult because of the unpredictable involvement of the urethral tissues.
    e. The defect is usually predictably proximal to the external sphincter at the junction of the prostatic urethra with the membranous urethra.
17. In determining the anatomy of the stricture, all of the following provide useful information EXCEPT:
    a. Magnetic resonance imaging (MRI).
    b. High-resolution ultrasonography.
    c. Contrast studies.
    d. Urethroscopy.
    e. Calibration with bougie à boule.
18. With regard to planning of reconstruction for urethral stricture, which of the following statements is TRUE?
    a. Even if a patient does not have retention, placement of a suprapubic tube may help define strictured areas.
    b. Tightly stenotic areas should be dilated to pass endoscopes proximally.
    c. The effects of hydrodilation are manifested most immediately distal to the area of narrowest stenosis.
    d. Calibration of strictured areas to 16 Fr or greater reliably predicts the potential for segments to contract.

e. Sonourethrogram by itself accurately predicts depth of spongiofibrosis.

19. With regard to direct visual internal urethrotomy, which of the following statements is TRUE?
    a. Strictures are best incised at the 12 o'clock position.
    b. Deep incision of the corpus spongiosum has been shown to optimize long-term results.
    c. In optimally selected patients, long-term success of internal urethrotomy is approximately 90%.
    d. Internal urethrotomy should be the first procedure considered for any stricture of the anterior urethra.
    e. It can be associated with erectile dysfunction (ED).
20. Concerning anterior urethral reconstruction, which of the following statements is TRUE?
    a. Excision and primary anastomosis reconstruction are severely limited and useful only for very proximal strictures 1 to 2 cm in length.
    b. Performance of the excision and primary anastomosis technique is facilitated by dissection of the corpus spongiosum to the level of the glans penis.
    c. Success requires total excision of the fibrosis with a widely spatulated anastomosis.
    d. Reconstruction is facilitated by development of the intracrural space with infrapubectomy.
    e. In cases of longer strictures, excision with partial anastomosis allowing one wall to granulate offers acceptable results.
21. With regard to genital skin flaps for anterior urethral reconstruction, which of the following statements is TRUE?
    a. Flap operations are best applied as individual techniques and require the surgeon to be intimately familiar with the individual steps of each technique.
    b. The operation can conceptually become one operation with multidimensional application.
    c. The operations are all based on mobilization of the extended Buck fascia.

d. The operations require a comfortable understanding of the extended circumflex iliac superficial vascular pattern.
e. They are of limited value in patients who have been previously circumcised.

22. With regard to continence after reconstruction for PFUIs, which of the following statements is TRUE?
    a. Location of the injury along the course of the membranous urethra is not associated with continence postoperatively.
    b. Continence can be accurately predicted by contrast studies.
    c. Continence is best predicted by the appearance of the bladder neck on endoscopy.
    d. Continence is best addressed after a procedure to reestablish urethral continuity is performed.
    e. Continence is best in patients with partial distraction injuries.
23. In dealing with the entity of chordee without hypospadias, which of the following statements is TRUE?
    a. Correction of curvature is often achieved with mobilization of the corpus spongiosum alone.
    b. It often can be corrected with maneuvers that lengthen the foreshortened ventral skin.
    c. It is best straightened by an incision and grafting operation.
    d. Division of the urethra/corpus spongiosum is virtually always indicated.
    e. It is usually present with either ventral curvature or ventral curvature associated with torsion.
24. With regard to acquired curvatures of the penis that are not Peyronie disease, which of the following statements is TRUE?
    a. Most are characterized by prominent dorsal scars.
    b. In most cases, global cavernosal veno-occlusive dysfunction (CVOD) is not a complicating factor.
    c. They are virtually never associated with “minimal” buckling trauma.
    d. Patients often have significant penile foreshortening.
    e. There is usually an association with either hypospadias or epispadias.
25. All of the following are true regarding PFUIs, EXCEPT:

a. Pelvic fractures are associated with urethral injuries in about 10% of cases.
b. PFUIs most commonly occur between the prostate and membranous urethra.
c. In the prepubescent male, PFUI's are more likely to involve the prostatic urethra.
d. A normal anterior urethral on retrograde urethrography nearly ensures an anastomotic repair of a PFUI is feasible.
e. The appearance of the bladder neck on preoperative contrast studies does not accurately predict continence outcomes following anastomotic repair of PFUIs.

26. In PFUIs, after excision of the traumatic scar, the distance between the two ends of healthy urethra can be minimized by all of the following, EXCEPT:
    a. mobilizing the corpus spongiosum off the corpora cavernosa up to the corona of the glans.
    b. excision of Buck fascia from the corpus spongiosum.
    c. dissection of the intracrural space down to the pubis.
    d. periosteal elevation and infrapubectomy.
    e. rerouting of the spongiosum above the crura of the corpora cavernosa.
27. Which of the following patients should be considered for penile revascularization to prevent ischemic stenosis following PFUI repair?
    a. Men with ED following PFUI, but normal hemodynamics on penile duplex sonography
    b. Men with normal erectile function following PFUI
    c. Men following PFUI with ED and hemodynamics on penile duplex sonography suggesting venous leak
    d. Men with arteriogenic ED following PFUI who demonstrate bilateral occlusion of the internal pudendal arteries without reconstitution
    e. Men with arteriogenic ED following PFUI who demonstrate unilateral occlusion of the internal pudendal artery
28. Which of the following statements regarding total phallic construction is FALSE?

a. Rigidity of the neophallus may be achieved by placement of a penile prosthesis before the return of tactile sensation of the phallus.
b. Current techniques are accomplished with a variety of flap designs, which use microvascular free flap transfer.
c. Urinary fistulae, although a common complication following phallic construction, are often resolved with conservative measures and do not routinely require operative repair.
d. Complications following prosthesis placement into patients following total phallic construction are higher than those following placement into men with normal corporal anatomy.
e. Penile tactile and erogenous sensation can be achieved following total phallic construction via coaptation of the flap cutaneous nerves to the dorsal penile/clitoral nerve, the pudendal nerve, or the ilioinguinal nerve.

29. All of the following may be repaired with excision and primary anastomotic urethroplasty, EXCEPT:
    a. PFUI with a 3-cm gap demonstrated on preoperative urethrography.
    b. straddle injury resulting in a 1-cm bulbar urethral stricture.
    c. iatrogenic urethral trauma during a transurethral resection resulting in a 1-cm bulbar urethral stricture.
    d. idiopathic proximal 1-cm bulbar urethral stricture in a patient with a history of a prior hypospadias repair.
    e. perineal trauma resulting in a 2-cm proximal bulbar urethral stricture.
30. Which of the following statements regarding vessel-sparing urethral reconstruction is FALSE?
    a. It preserves the vascularity of the bulbospongiosum.
    b. It requires a deep understanding of the anatomy and blood supply of the penile structures.
    c. It may be technically challenging in less experienced surgeons.
    d. By preserving the proximal blood supply of the urethra, it has better outcomes and less de novo erectile disfunction than a transecting excision and primary anastomosis.

e. There are different applications of the technique, both for anterior urethral reconstruction and posterior urethral reconstruction.

31. Which of the following are the best candidates for vessel-sparing urethral reconstruction?
    a. Young man with less than 2-cm proximal bulbar urethral stricture
    b. Prior failed bulbar urethral reconstruction
    c. Proximal bulbar urethral stricture in previously reconstructed hypospadias
    da, b, and c
    e. b and c

Answers

1. **b. A graft is tissue excised from a donor site that reestablishes its blood supply by revascularization.** The term *graft* implies that tissue has been excised and transferred to a graft host bed, where a new blood supply develops by a process termed *take.* Take requires approximately 96 hours and occurs in two phases. The initial phase, imbibition, requires about 48 hours. During that phase, the graft survives by "drinking" nutrients from the adjacent graft host bed, and the temperature of the graft is less than the core body temperature. The second phase, inosculation, also requires about 48 hours and is the phase in which true microcirculation is reestablished in the graft. During that phase, the temperature of the graft rises to core body temperature. The process of take is influenced by both the nature of the grafted tissue and the conditions of the graft host bed. Processes that interfere with the vascularity of the graft host bed thus interfere with graft take.
2. **b. The subdermal plexus is carried at the juncture of the deep dermis and the underlying tissue.** The epidermal, or epithelial layer, is a covering, the barrier to the "outside," and is adjacent to the superficial dermis, or superficial lamina. At approximately that interface is the superficial plexus. In the case of skin, the plexus is

the intradermal plexus. There are some lymphatics in the superficial dermal or tunica layer. On the undersurface of the deep dermal layer or deep lamina is the deep plexus. In the case of skin, this is the subdermal plexus. **The deep dermis contains most of the lymphatics and greater collagen content than found in the superficial dermal layer.** The deep or reticular dermis is generally thought to account for the physical characteristics of the tissue. There is a difference between genital full-thickness skin (penile and preputial skin grafts) and extragenital full-thickness skin. This is probably a reflection of the increased mass of the graft in extragenital skin grafts. This increased mass makes the graft more fastidious, and the poor results reported with urethral reconstruction with extragenital FTSGs are probably due to poor or ischemic take.

3. **c. Buccal mucosa graft is thought to have a panlaminar plexus.** In the bladder epithelial graft, there is a superficial and a deep plexus; however, many more perforators connect the plexuses. Thus bladder epithelial grafts tend to have more favorable vascular characteristics. In the case of the oral mucosal grafts, there is a panlaminar plexus. Thus the oral mucosal graft can be thinned somewhat, provided a sufficient amount of deep lamina is carried to preserve the physical characteristics. The oral mucosal grafts are thought to have optimal vascular characteristics. The thinned graft diminishes the total graft mass while preserving the physical characteristics and not adversely affecting the vascular characteristics. Tunica vaginalis grafts have been tried for urethral reconstruction with uniformly poor results. The dermal graft (not full-thickness skin) has been used for years to augment the tunica albuginea of the corpora cavernosa.
4. **b. The erectile tissues of the normal corpora cavernosa are separated from the tunica by the space of Smith. The corpora cavernosa are not separate structures but constitute a single space with free communication through an incompetent midline septum, composed of multiple strands of elastic tissue similar to that making up the tunica albuginea.** The erectile tissue is separated from the tunica albuginea by a thin layer of areolar connective tissue that was described by Smith. The Buck fascia is

directly abutted to the tunica albuginea of the corpora cavernosa. The Buck fascia surrounds the adventitia of the corpora spongiosum in envelope fashion, and the dorsal neurovascular structures are contained in envelope fashion between the superficial and deep laminar of the Buck fascia on the dorsum. The Buck fascia is thus "devoted" to the deep structures. The dartos fascia is loosely areolar and lies immediately beneath the skin. It is in that fascial layer that the arborizations of the superficial external pudendal vessels and the posterior scrotal vessels are carried.

5. **b. The bulbous urethral portion is invested by the thickest portion of the corpora spongiosum.** The fossa navicularis is contained within the spongy erectile tissue of the glans penis and terminates at the junction of the urethral epithelium with the skin of the glans. The bulbous urethra is covered by the midline fusion of the ischiocavernosus musculature and is invested by the bulbospongiosum of the proximal corpus spongiosum. It becomes larger and lies closer to the dorsal aspect of the corpus spongiosum, exiting from its dorsal sur face before the posterior attachment of the bulbospongiosum to the perineal body. The membranous urethra is the portion that traverses the perineal pouch and is partially surrounded by the external urethral sphincter. This segment of the urethra is unmatched to fixed structures, has the distinction of being the only portion of the male urethra that is not invested by another structure, and is lined with a delicate transitional epithelium.
6. **e. The blood supply of the deep structures of the penis is derived from the common penile artery, which is the continuation of the internal pudendal artery after it branches off its perineal branch and the posterior scrotal arteries.** The common penile artery is the continuation of the internal pudendal artery giving off perineal posterior scrotal branches. From that point onward, it is termed the *common penile artery.* As it nears the urethral bulb, the artery divides into its three terminal branches as follows: (1) the bulbourethral arteries, which enter the proximal corpus spongiosum; (2) the dorsal artery, which travels along the dorsum of the penis contained in envelope fashion between the superficial

and deep lamina of the Buck fascia; and (3) the cavernosal arteries, usually a single artery, which arise and penetrate the corpora cavernosa at the hilum and run the length of the penile shaft. The circumflex cavernosal arteries are given off at varying locations along the dorsal artery, but their distribution is neither uniform nor dependable.

7. **c. The dorsal nerve arises in the Alcock canal as a branch of the pudendal nerve.** The cavernosal nerves are a combination of the parasympathetic and visceral afferent fibers that constitute the autonomic nerves of the penis. These provide the nerve supply to the erectile apparatus. The pudendal nerves enter the perineum with the internal pudendal vessels through the lesser sciatic notch at the posterior border of the ischiorectal fossa. They run in the fibrofascial pudendal canal of Alcock to the edge of the urogenital diaphragm. Each dorsal nerve of the penis arises on Alcock canal as the first branch of the pudendal nerve. On the shaft, their fascicles fan out to supply proprioceptive and sensory nerve terminals in the tunica of the corpora cavernosa and sensory terminals in the skin. The skin of the penis is innervated by branches of the genitofemoral nerve.
8. **c. There is a strong association with an infectious etiology for the development of LS.** The etiology of LS has not been well defined. Multiple potential infectious etiologies have been suggested, but recent studies have found no association. Other proposed etiologies include a Koebner phenomenon, autoimmune event, and genetic associations. The remaining answer choices are correct.
9. **b. Genital or extragenital skin may be used for urethral reconstruction for LS-associated urethral strictures. Most surgeons now believe that LS is a disease of genital skin. For this reason, genital skin is not appropriate for reconstruction in patients with LS. Although it is technically possible to use extragenital skin for reconstruction, oral mucosal grafting has emerged as a better tissue in patients with LS associated urethral strictures.** Patients with meatal stenosis from LS not infrequently also have more proximal strictures and need a

complete workup prior to surgery. Patients with long-standing LS should be monitored for potential development of squamous cell carcinoma, because this has been reported.

10. **d. Complication of genital herpes simplex viral infection.** Genital HSV has not been reported to cause urethrocutaneous fistulae. Each of the remaining answers is a potential etiology for urethrocutaneous fistula formation.
11. **c. When ammoniacal meatitis is noted, often a short course of meatal dilation and steroid cream application will resolve the problem. Meatal stenosis in the male child appears to be a consequence of circumcision, which allows for ammoniacal meatitis.** Children seen with ammoniacal meatitis are usually started with meatal dilation using steroid cream. Within a week, the process seems to settle down. Anecdotally, the fusion of the ventral meatus skin, which causes meatal stenosis, seems to be avoided. Because childhood meatal stenosis truly represents a fusion of the ventral urethral meatus, dividing the thin membrane of fusion is preferred. This leaves the child with a slit-shaped meatus.
12. **e. The McRoberts technique of macroreplantation is not the preferred method of management for these patients, but when the situation warrants it, it is very successful.** Often the amputation is self-inflicted, usually during an acute psychotic break. This should not preclude replantation unless the patient adamantly refuses such treatment. Even then, with a court order and the agreement of two or more surgeons, replantation may be undertaken. If possible, microreplantation should be carried out. This technique consists of an anatomic approximation of the tunica albuginea of the corporal bodies, a spatulated two-layer anastomosis of the urethra. The dorsal nerves are coapted using an epineural technique unless the injury is distal, at which point a vesicular coaptation may be required. The dorsal vein is anastomosed, and the dorsal arteries are anastomosed. Anastomosis of the cavernosal arteries is not possible and should not be attempted. If the situation is such that microreplantation cannot be undertaken, then the technique described by McRoberts can be carried out. His series and other series show that a high degree of

success can be expected after replantation without microvascular reanastomosis. In most patients, however, they will have numbness distal to the replant site. With microreplantation, it is not at all unusual for patients to have excellent sensation distal to the area of injury and to have resumption of normal erectile function.

If the patient presents with the distal part having been disposed of or otherwise unavailable, then the wound should be closed. Often the penis will have been stretched during the amputation and an excess of skin will have been removed, leaving a good length intact with denuded penile shaft structures. In that case, the corporeal bodies would be closed, the urethral meatus must be spatulated, and the penis can be immediately covered with an STSG. If the injury occurs because of avulsion, replantation is not an option as the stretch injury to the spermatic vessels or vessels of the penis causes unpredictable damage to the endothelium.

13. **a. Circumcision may be performed in the neonatal period in newborns born with a distal hypospadias, but not in those with proximal hypospadias. Newborns with any concern for hypospadias (distal or proximal) should NOT undergo circumcision.** It is essential to preserve the foreskin so that it may be used for the hypospadias repair if necessary. **Circumcision has consistently been shown in well-conducted randomized controlled trials to reduce the risk of HIV acquisition in heterosexual men by 50% to 60%. Circumcision will reduce the risk for UTI in infants and should be considered in those with recurrent infections.** Most skin dehiscence following circumcision can be managed conservatively and does not require operative intervention.
14. **a. Reconstruction for lymphedema that is the consequence of the indirect effects of radiation is best accomplished with excision of the tissues and coverage with STSGs.** Patients with lymphedema can readily undergo reconstruction. When the lymphedematous tissue has been excised, the testes will be free and, as in a degloving injury, they must be fixed in the midline in an anatomically correct position. The shaft of the penis should be covered with a STSG. If the scrotum cannot be closed, a meshed

STSG is used to cover the testes, as described. Not uncommonly, these patients have hydroceles, the parietal tunica vaginalis must be excised, and grafting can be done directly onto the visceral tunica vaginalis of the testicles. **Unlike the full-thickness skin flap (FTSF), split-thickness skin carries little of the reticular dermis and hence few of the lymphatic channels.** Reaccumulation of lymphedema will occur within a FTSG and can recur in a thick STSG. In many cases of lymphedema limited to the genitalia, the posterior scrotal skin and the lateral scrotal skin are spared from the lymphedematous process. Thus, in some cases, primary closure after excision can be accomplished using these tissues. If grafting is required, using these tissues to blend the grafts into the groin and perineum technically is much easier. The lymphedematous process involves recurrent cellulitis, lymphedema, and the development of lymphangiectasia. Lymphangiectasia can look like genital papilloma; however, it is a very different process. If there is any question, biopsy can clarify the issue.

15. **d. It causes limitation of the urethral lumen because of contraction and noncompliance of the scar.** The term *urethral stricture* refers to anterior urethral disease. By virtue of the Consensus Conference, obliterative processes of the membranous urethra, such as those associated with pelvic fracture, would be referred to as PFUI, and other narrowing processes of the posterior urethra are correctly referred to as either contractures or stenosis. **Thus the term *urethral stricture* describes a process that involves the urethral epithelium along with the spongy erectile tissue of the corpus spongiosum, and this is referred to as spongiofibrosis.** In some cases, the scarring process can extend through the tissues of the corpus spongiosum and into the adjacent tissues. It is contraction of the scar that reduces the urethral lumen. Squamous metaplasia is often seen involving the urothelium of the urethra proximal to a narrow caliber urethral stricture.
16. **c. It is not a true stricture but rather fibrosis that results from distraction of the urethra.** By virtue of the Consensus Conference, narrowing of the posterior urethra is not referred to as a stricture. Those obliterative processes associated with pelvic

fracture are termed PFUI. PFUI is an obliterative process of the posterior urethra that has resulted in fibrosis and is the defect of distraction of the urethra in that area. Although the distraction defect can be lengthy in some cases, the actual process involving the tissues of the urethra is usually confined.

17. **a. Magnetic resonance imaging (MRI).** To devise an appropriate treatment plan, it is important to determine the location, length, depth, and density of the stricture (spongiofibrosis). The length and location of the stricture can be determined using radiographs, urethroscopy, and ultrasonography. The depth and density of the scar in the spongy tissue can be deduced from the physical examination, the appearance of the urethra in contrast studies, the amount of elasticity noted on urethroscopy, and the depth and density of fibrosis as evidenced by ultrasonographic evaluation of the urethra, although the absolute length of spongiofibrosis may not be evident on ultrasonographic evaluation. MRI has been suggested as useful in patients with pelvic fracture urethral distraction, particularly in cases in which the anatomy of the pelvis has become significantly distorted. With regard to anterior urethral stricture, however, MRI has not been useful, with the exception of those cases in which there is urethral carcinoma. In those cases, MRI can provide invaluable information concerning the spread of the tumor. Bougie à boule calibration can be very helpful.
18. **a. Even if a patient does not have retention, placement of a suprapubic tube may help define strictured areas.** In selected patients, we have found it useful to place a suprapubic tube to defunctionalize the urethra. After 6 to 8 weeks, if there will be a constriction of an area that was hydrodilated with voiding, the tendency for that constriction to occur should become apparent. **It is imperative, however, to completely evaluate the urethra proximal and distal to the stricture with endoscopy and bougienage during surgery, to ensure that all of the involved urethra is included in the reconstruction.** Whereas hydraulic pressure generated by voiding may keep segments proximal to the stricture patent, unless these segments are included in the repair, they are at risk for contraction after obstruction of the narrow-

caliber segment is relieved with reconstruction. For this reason, any abnormal areas of the urethra that are proximal to a narrow-caliber segment of the stricture must be treated with suspicion. If the lumen does not appear to demonstrate evidence of diminished compliance, then we presume that area to be uninvolved in active stricture disease. However, coning down of the urethra suggests its involvement in the scar. Use of a sonourethrogram is thought by some to accurately establish length of stricture but not the extent of spongiofibrosis.

19. **e. It can be associated with ED.** Many surgeons have learned to perform internal urethrotomy by making a single incision at the 12-o'clock position. This location might be questioned, however, based on the location of the urethra within the corpus spongiosum. Distally, although the anterior aspect of the corpus spongiosum is thicker, a deep incision in the more distal aspects of the anterior urethra will certainly enter the corpora cavernosa, and these incisions have been associated with the creation of ED. The most common complication of internal urethrotomy is recurrence of stricture. Less commonly noted complications of internal urethrotomy include bleeding and extravasation of irrigation fluid into the perispongiosal tissues. **One report that used the actuarial technique showed the curative success rate of internal urethrotomy to be 29% to 30% for all patients. Other evaluations have confirmed this success rate. However, a number of studies do show which strictures best respond to internal urethrotomy. These are strictures of the bulbous urethra that are less than 1.5 cm in length and are not associated with the dense or deep spongiofibrosis (i.e., straddle injuries).** In those particular cases, long-term success has been shown to be 75% to 78%. For strictures outside the bulbous urethra, most studies do not show internal urethrotomy to have long-term success.
20. **c. Success requires total excision of the fibrosis with a widely spatulated anastomosis. It has now been demonstrated with certainty that the most dependable technique of anterior urethral reconstruction is the complete excision of the area of**

fibrosis, with a primary reanastomosis of the normal ends of the anterior urethra. The best results are achieved when the following technical points are observed: (1) the area of the fibrosis is totally excised; (2) the urethral anastomosis is widely spatulated, creating a large ovoid anastomosis; (3) the anastomosis is tension free; (4) epithelial apposition is achieved. With vigorous mobilization, development of the intracrural space, and detachment of the bulbospongiosum from the perineal body, significant lengths of stricture can be excised and reanastomosed. For very proximal bulbous strictures, tension-free anastomosis can be facilitated by the dissection of the membranous urethra. As a rule, the closer the stricture is to the membranous urethra, the longer it can be and still be reconstructed by anastomotic techniques. The tenet that excision and primary anastomosis should be the goal for all bulbous strictures is one that is being further reinforced by current published series. Although guideline lengths of 1 to 2 cm are valuable for planning, most would agree that if excision and primary anastomosis is possible it should be done, and, with aggressive dissection and the maneuvers described earlier, often strictures much longer than the "guideline lengths" can be so reconstructed.

21. **b. The operation can conceptually become one operation with multidimensional application.** A number of applications of genital skin islands, mobilized on either the dartos fascia of the penis or the tunica dartos of the scrotum, have been proposed for repair of urethral stricture disease. In the past, these flap operations were considered to be separate procedures. We suggest that all of these procedures are really different applications of a single concept, proposed by the microinjection studies of Quartey. Skin islands, as mentioned, can be viewed as passengers on fascial flaps, and the design of flaps for urethral reconstruction can be done parallel to the design of flaps for reconstruction in general. These procedures that use skin islands oriented on the penile dartos fascia have been also useful for reconstruction of the fossa navicularis. There are three important considerations for the use of flaps in urethral reconstruction: (1) the nature of the flap tissue, (2) the

vasculature of the flap, and (3) the mechanics of flap transfer. The skin must be nonhirsute for urethral reconstruction. In addition, for donor site consideration, it is most convenient to use the areas of redundant nonhirsute genital skin. These skin islands can be reliably elevated even in patients who have been circumcised.

22. **d. Continence is best addressed after a procedure to reestablish urethral continuity is performed.** We have found, and others have reported, that the competence of the bladder neck is difficult to accurately assess before the reestablishment of urethral continuity. Even in cases in which an obvious scar is noted to involve the bladder neck, follow-up of these patients after urethral reconstruction has found many patients with more than adequate continence. Still other patients are believed to have incontinence due to scar incarceration of the bladder neck. In our experience, however, this is an infrequent occurrence, and the appearance of the bladder neck by any modality available is not predictive of continence. It is currently our practice to reestablish the continuity of the urethra and, in cases in which there are concerns about continence, to forewarn the patient before the urethral reconstruction. Colopinto and others have not shown an association of ultimate continence as related to the location of the distraction injury.
23. **e. It is usually present with either ventral curvature or ventral curvature associated with torsion. In many cases, there are abnormalities of the ventral penile skin.** In patients who have chordee without hypospadias, the photograph will reveal an erect penis commensurate with the size of the detumescent penis, whereas in the congenital curvature patient, the erect penis will be noticeably large. Because of their congenital anomaly, these patients often become relatively reclusive and have poor self and genital images and may benefit from sex therapy. Even in patients with obvious abnormalities of the corpus spongiosum (i.e., poor ventral fusion or frank bifid corpus spongiosum), wide mobilization usually reveals that it is not the corpus spongiosum that remains as the ventral limiting factor. **In most patients, the penis will remain curved because of the inelasticity of the**

ventral aspect of the corpora cavernosa. If the epithelial tube has served as an adequate urethra (i.e., it is not stenotic), the morbidity of the urethral division and subsequent need for urethral reconstruction must be considered before undertaking such a procedure. In children, after mobilization and excision of the dysgenetic tissues, the residual chordee can usually be corrected by making a longitudinal incision with a sharp blade. If this maneuver is not sufficient, the dorsal neurovascular structures can be mobilized in concert with the Buck fascia and a small ellipse or ellipses of dorsal tunica albuginea excised and closed with watertight plicating sutures.

24. **b. In most cases, global cavernosal veno-occlusive dysfunction (CVOD) is not a complicating factor.** When a young man presents with an acquired curvature of the penis, one must always allow for the possibility of Peyronie disease. Occasionally, however, a patient or his initial-care physician will ignore the stigmata of the trauma (often described as "minimal" by patients), and the patient will present with a noticeable lateral scar that causes both indentation of the lateral aspect of the penis and, in some cases, curvature. Patients who had preexisting lateral curvature may actually notice that their penis has been straightened by the trauma, but they are disturbed by the concavity caused by the scar. The pathology of a subclinical fracture of the penis is believed to be due to either the disruption of the outer longitudinal layer of the tunica albuginea during the buckling trauma only or the disruption of both layers of the tunica albuginea during the buckling trauma but with preservation of Buck fascia. These patients usually have normal erectile function, and there is no association with concomitant global CVOD. However, the association of CVOD and trauma of the penis continues to be seen, and some patients, after fracture-type injuries of the penis, will have significant problems with ED. These injuries are not associated with shortening of the penis. It is the lack of ED and penile shortening that help distinguish these patients from those with Peyronie disease. Although foreshortening of the penis is not a characteristic of either the injury itself or the resulting scar in either of these

injuries, these patients are not thought to be best treated by approaching the opposite aspect of the scar and excising an ellipse of the tunica. This would result in bilateral scars, which will cause bilateral indentation of the penis, and although the penis will have been straightened by the correction, most patients are upset by the cosmetic and functional result of a near-circumferential indentation of the penis. Curvatures associated with hypospadias or epispadias are not acquired curvatures.

25. **b. PFUIs most commonly occur between the prostate and membranous urethra.** Pelvic fracture distraction injuries of the membranous urethra have been compared with plucking an apple (prostate) off its stem (the membranous urethra). This analogy implies that the injury most frequently occurs at the apex of the prostate. **Experience shows that this is not the case, however, and the most frequent point of distraction is at the departure of the bulbous urethra from the membranous urethra. The distraction can, however, involve all or any portion of the membranous urethra between the departure of the bulbous urethra and the apex of the prostate.** The remaining answer choices are correct.
26. **a. Mobilizing the corpus spongiosum off the corpora cavernosa up to the corona of the glans. Aggressive mobilization of the corpus spongiosum is performed with caution because it is thought to have possible ill effects on retrograde blood supply, which in the pelvic fracture patient may be tenuous.** Meticulous detachment of the investment of Buck fascia from the corpus spongiosum increases the compliance of the corpus and limits the need for aggressive mobilization. It is important to try to avoid the creation of chordee during the repair of a distraction injury. To prevent chordee, the attachment cannot be carried beyond the area of the penoscrotal attachment. Development of the intracrural space, infrapubectomy, and, if needed, rerouting of the corpus spongiosum each shorten the course that the corpus spongiosum must traverse and allow reconstruction without attendant chordee.
27. **d. Men with arteriogenic ED following PFUI who demonstrate bilateral occlusion of the internal pudendal arteries without**

reconstitution. Patients with an intact pudendal artery on one side are often potent and reliably cured with reconstruction. Patients with only reconstituted vessels, either unilateral or bilateral, are rarely potent but reliably reconstructed. Those patients at risk for ischemic stenosis are only those with bilateral complete obstruction of the internal pudendal vessels without reconstitution. In such a patient, we now perform penile arterial revascularization to augment the vascularity and, with that accomplished, then proceed to urethral reconstruction. Patients without ED by definition have normal penile hemodynamics and do not require further investigation before repair of a PFUI.

28. **a. Rigidity of the neophallus may be achieved by placement of a penile prosthesis before the return of tactile sensation of the phallus.** Rigidity for intercourse in the patient with phallic construction is usually achieved by either an externally applied or a permanently implanted prosthesis. Prosthetic implantation is never undertaken until 1 year after phallic construction, because protective sensibility must be demonstrated in the flap. When the flap is transferred, it is, by definition, rendered insensate. At about 3 to 4 months after reconstruction, however, as nerve regeneration occurs, sensation becomes noticeable. In addition, before prosthetic implantation is undertaken, the urethra must be patent and proved to be durable.
29. **d. Idiopathic proximal 1-cm bulbar urethral stricture in a patient with a history of a prior hypospadias repair.** Patients with a history of hypospadias can be expected to have altered or absent retrograde blood supply to the urethra through the normal arborization in the glans. In this situation, transecting techniques to repair urethral strictures should be avoided unless performed in a "vessel-sparing" fashion to avoid the risk of ischemic urethral stenosis. Inlay or onlay graft or flap techniques may be used in these patients, as these do not disrupt the proximal blood supply to the urethra. The remaining scenarios can be reliably repaired with anastomotic urethroplasty.
30. **d. By preserving the proximal blood supply of the urethra, it has better outcomes and less de novo erectile disfunction a than**

transecting excision and primary anastomosis. The vessel-sparing technique for anterior urethral reconstruction was first introduced by Jordan et al. in 2007 and later modified by Mundy et al. in 2010. It spares the bulbar urethral arteries but maintains the same principles of traditional excision and primary anastomosis (wide, spatulated, tension-free anastomosis). Originally intended to prevent potential future spongiosum atrophy in selected patients, it became part of the armamentarium in anterior urethral reconstruction for a variety of entities in some centers around the world, although it may be reserved for experienced urethral reconstructive surgeons. To date, there is no evidence of benefit in terms of overall outcome or prevention of de novo ED. Since its introduction, the principle of vessel-sparing has been applied in a variety of techniques for repair of the anterior and even posterior urethra.

31. **e. b and c.** Despite that vessel-sparing urethral reconstruction has been embraced in several centers across the globe, there is no evidence that this technique is better than traditional excision and primary anastomosis. In cases where the proximal blood supply has already been compromised, such as stricture recurrence after prior transecting bulbar urethroplasty, or where the distal urethral blood supply of the urethra is poor, such in patients with hypospadias, especially after prior repairs, it makes sense to preserve the blood supply of the bulbospongiosum.

Chapter Review

1. A meshed split-thickness graft that is applied to the genitalia should not be expanded but rather placed on the recipient site without expansion to allow collections from beneath the graft to escape.
2. Split-thickness grafts may contain some lymphatics; however, full-thickness grafts have a full complement of lymphatics.
3. Split-thickness grafts are more likely to take (become vascularized) but tend to contract and become brittle when mature, whereas full-

thickness grafts have more difficulty becoming vascularized but are less likely to contract and are more durable when mature.

4. Tunica vaginalis grafts result in aneurysmal dilatation when they are used for large defects.
5. The superficial dorsal penile vein usually drains to the left saphenous vein; the deep dorsal and circumflex veins lying beneath the Buck fascia drain to the periprostatic plexus.
6. The Buck fascia is adjacent to the deep structures of the penis; the dartos fascia is next to the skin.
7. LS is a disease of the skin and may involve large portions of the genital skin; therefore, using the genital skin for reconstruction may result in recurrence of the disease. Oral mucosal grafting has emerged as a better tissue in patients with LS–associated urethral strictures.
8. LS may be a premalignant lesion and often results in meatal stenosis.
9. A spontaneous urethral fistula or unexplained periurethral abscess may be the harbinger of a urethral carcinoma.
10. Circumcision provides protection for heterosexual men in areas of endemic HIV; it reduces the risk of acquiring herpes simplex type 2, papillomavirus, and genital ulcer disease.
11. Cellulitis may be a problem in patients who have genital lymphedema.
12. As a general rule in the urethra, flaps are best suited for distal reconstruction, grafts for proximal reconstruction.
13. A urethral stricture involves the epithelium as well as the corpus spongiosum (spongiofibrosis).
14. For urethral distraction injuries (posterior urethral disruptions), an aligning catheter, at the very worst, facilitates subsequent reconstruction and, at best, may leave the patient with an endoscopically manageable urethra.
15. Because paraphimosis tends to recur, a dorsal slit or circumcision should be electively planned.
16. For urethral reconstruction an onlay graft or flap has a higher success rate than tabularized grafts.
17. Vesicourethrorectal fistulae are most successfully closed when normal tissue is interposed between the rectum and the

bladder/urethra. The gracilis muscle interposition flap is an excellent tissue to interpose and, when used, has a high success rate.

18. Curvature of the penis in patients with Peyronie disease who require repair are best managed with corporal plication techniques and not grafting.
19. The corpora cavernosa are not separate structures but constitute a single space with free communication through an incompetent midline septum.
20. Meatal stenosis in the male child appears to be a consequence of circumcision.
21. It is imperative to completely evaluate the urethra proximal and distal to a stricture with endoscopy and bougienage during surgery, to ensure that all of the involved urethra is included in the reconstruction.
22. The curative success rate of internal urethrotomy is 29% to 30% for all patients. The strictures that respond best to internal urethrotomy are strictures of the bulbous urethra that are less than 1.5 cm in length and are not associated with the dense or deep spongiofibrosis.
23. The best results for primary reanastomosis are achieved when the following technical points are observed: (1) the area of the fibrosis is totally excised; (2) the urethral anastomosis is widely spatulated, creating a large ovoid anastomosis; (3) the anastomosis is tension free; and (4) epithelial apposition is achieved.
24. In complete membranous urethra disruption, the most frequent point of distraction is at the departure of the bulbous urethra from the membranous urethra. The distraction can, however, involve all or any portion of the membranous urethra between the departure of the bulbous urethra and the apex of the prostate.
25. Aggressive mobilization of the corpus spongiosum is performed with caution in patients with urethral disruptions because it is thought to have possible ill effects on retrograde blood supply to the urethra.

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Surgery of the Scrotum and Seminal Vesicles

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Keywords

chronic pain; hydrocele; microscopic varicocelectomy; scrotum; seminal vesicle cyst; seminal vesicles/surgery; seminal vesicles; surgical procedures; vasectomy

Questions

1. Which of the following vessels has the least direct contribution to the arterial supply of the vas deferens?
    a. Deferential artery
    b. Internal spermatic artery
    c. Superior vesicle artery
    d. Inferior epigastric artery
    e. Inferior epididymal artery
2. The best reason for using the no-scalpel vasectomy technique is:
    a. it has a higher sterilization rate than standard vasectomy with incision.
    b. patients are rendered sterile in less time.
    c. it is easier to learn than the standard technique.
    d. it results in a lower rate of complications, including hematoma and infection.
    e. it results in a higher rate of reversibility.

3. The no-scalpel technique for vasectomy reduces the rate of:
    a. hematoma.
    b. vasectomy failures.
    c. recanalization.
    d. injury to testicular artery.
    e. chronic orchialgia.
4. Vasectomy failure rate when both the abdominal and testicular ends of the divided vas deferens are occluded with hemoclips is:
    a. less than 1%.
    b. 5% to 10%.
    c. 10% to 20%.
    d. 20% to 30%.
    e. 50% to 60%.
5. The technical aspect shown to decrease vasectomy failure rates the most is:
    a. no-scalpel technique.
    b. conventional technique.
    c. fascial interposition of dartos fascia between the divided ends of the vas deferens.
    d. occluding both ends of the divided vas deferens with hemoclips.
    e. occluding both ends of the divided vas deferens thermally with the use of intraluminal cautery.
6. The technical aspect when performing vasectomy to make vasectomy reversal easier in the future is:
    a. no-scalpel technique.
    b. not excising a long segment of vas deferens.
    c. dividing the vas deferens as close to the epididymis as possible.
    d. occluding both ends of the divided vas deferens with hemoclips.
    e. occluding both ends of the divided vas deferens thermally with the use of intraluminal cautery.
7. Vasectomy has been established as associated with:
    a. prostate cancer.
    b. dementia.

c. cardiovascular disease.
d. atherosclerosis.
e. a 10% incidence of chronic scrotal pain.

8. What is the estimated percentage of men who develop antisperm antibodies after vasectomy?
   a. 0% to 20%
   b. 20% to 40%
   c. 40% to 60%
   d. 60% to 80%
   e. Greater than 80%
9. Which of the following is an indication for repeat vasectomy?
   a. Painless sperm granuloma
   b. Motile sperm found in semen analysis 3 months after vasectomy
   c. Nonmotile sperm found in semen analysis 3 months after vasectomy
   d. Persistent testicular pain 3 months after vasectomy
   e. All of the above
10. Pressure-induced injury following vasectomy occurs in:
   a. the testis.
   b. the ejaculatory duct.
   c. the epididymis.
   d. the vas deferens.
   e. the seminal vesicles.
11. In the management of chronic orchialgia, which of the following statements is TRUE?
   a. Imaging studies are not indicated.
   b. Varicocele is not a significant contributor of chronic scrotal pain.
   c. Orchiectomy usually relieves the pain.
   d. Denervation of the cord may offer relief in selected cases.
   e. Diagnostic epididymal puncture should be performed to rule out chronic bacterial epididymitis.
12. Which of the following statements is TRUE regarding hydrocelectomy?
   a. Hematoma is the least frequent complication.

b. The Jaboulay bottleneck operation is associated with a high recurrence rate.
c. The Lord plication is an ideal operation for long-standing postinfectious hydroceles.
d. Sclerotherapy is often the treatment of choice for young men of reproductive age.
e. The Jaboulay bottleneck operation is associated with a low recurrence rate.

13. A non-transilluminating, nontender mass is noted in the epididymis on physical examination and confirmed to be solid by sonography. What is the most likely diagnosis?
    a. Epididymal cyst
    b. Adenomatoid tumor
    c. Spermatocele
    d. Testicular tumor
    e. Hydrocele
14. Men who were treated with epididymectomy for chronic epididymitis responded the most favorably if:
    a. there was a palpable epididymal abnormality.
    b. there was no palpable abnormality, but there were sonographic changes of the epididymis.
    c. there were no palpable abnormalities and no sonographic changes of the epididymis.
    d. they had improvement of pain with spermatic cord block.
    e. none of the above apply.
15. Which of the following statements is TRUE regarding retractile testes in adults?
    a. As in children, surgical repair is never indicated.
    b. A dartos pouch orchidopexy is the treatment of choice.
    c. Simple three-stitch orchiopexy of the tunica albuginea to the dartos, as for torsion prophylaxis, is effective in preventing retraction.
    d. Bilateral orchiopexy is necessary for a unilateral retractile testis.
    e. Coexisting varicocele is common.

16. The most appropriate approach to a long-standing, thick-walled, loculated hydrocele is:
    a. excision of the hydrocele sac.
    b. the Jaboulay bottleneck technique.
    c. the Lord plication technique.
    d. the inguinal approach.
    e. sclerotherapy.
17. In men with chronic orchitis without an identifiable bacterial pathogen, antibiotics:
    a. decrease the length of symptoms.
    b. improve the severity of symptoms.
    c. decrease the length of time to full activity.
    d. are steadily being prescribed more frequently empirically.
    e. none of the above apply.
18. When a clinically palpable varicocele is encountered in a patient with orchialgia, varicocelectomy will resolve the pain:
    a. 10% of the time.
    b. 25% of the time.
    c. 50% of the time.
    d. 75% of the time.
    e. 90% of the time.
19. What is the embryologic origin of the seminal vesicles?
    a. Müllerian duct
    b. Ectodermal ridge
    c. Distal mesonephric duct
    d. Swelling of the distal paramesonephric duct
    e. Neural crest cells
20. What percentage of the ejaculate volume is made up of seminal vesicle secretions?
    a. 5% to 10%
    b. 20% to 30%
    c. 60% to 80%
    e. 90%
    e. The seminal vesicle does not contribute to the seminal plasma volume
21. What artery is the major blood supply to the seminal vesicle?

a. Hypogastric
b. Vesiculodeferential artery
c. Inferior vesicle
d. Internal iliac
e. Deep dorsal penile

22. Decreased T1 signal intensity on MRI, along with increased T2 intensity of seminal vesicles, is indicative of which process?
    a. Inflammation of the seminal vesicles
    b. Hemorrhage within the seminal vesicles
    c. Seminal vesicle tumors
    d. Seminal vesicle cysts
    e. Normal seminal vesicles
23. Agenesis of the seminal vesicle is associated with significant ipsilateral renal anomalies. What is the embryologic reason for this?
    a. A genetic defect links seminal vesicle agenesis to renal agenesis.
    b. A mutation occurs in the cystic fibrosis transmembrane regulator gene.
    c. There was an insult to the mesonephric duct at approximately 12 weeks' gestation.
    d. There was an embryologic insult to the mesonephric duct earlier than 7 weeks' gestation.
    e. There is no association between agenesis of the seminal vesicle and ipsilateral renal anomalies.
24. What disorder is frequently associated with bilateral agenesis of the seminal vesicles?
    a. Cystic fibrosis
    b. Kartagener syndrome
    c. Young syndrome
    d. Kallmann syndrome
    e. Klinefelter syndrome
25. What causes the majority of seminal vesicle cysts?
    a. Ejaculatory duct stone
    b. Obstruction of the ejaculatory duct
    c. Inflammation

d. Renal agenesis
e. Trisomy 21

26. What is the most common type of malignant neoplasm found in seminal vesicles?
    a. Primary adenocarcinoma
    b. Sarcoma
    c. Cystosarcoma phyllodes
    d. Metastatic tumors
    e. Amyloidosis
27. What is the best initial test for a suspected seminal vesicle abnormality?
    a. Computed tomography (CT)
    b. Transrectal ultrasonography
    c. Magnetic resonance imaging (MRI)
    d. Fine-needle biopsy
    e. Vasography
28. What is the best method to differentiate a benign from malignant seminal vesicle mass?
    a. Biopsy of the lesion
    b. Contrast medium-enhanced CT
    c. Gadolinium-enhanced MRI
    d. Transrectal ultrasonography
    e. Rectal examination
29. What is the best surgical approach to a congenital lesion of the seminal vesicle?
    a. The perineal route because this has the quickest recovery.
    b. The transcoccygeal route because these are usually large lesions.
    c. The laparoscopic route so that the ipsilateral kidney can be dealt with concomitantly and recovery may be shorter.
    d. The paravesical route because this has a lower incidence of postoperative erectile dysfunction.
    e. The transvesical route because rectal injury is much less likely.
30. What is the best indication for the transcoccygeal approach to the seminal vesicle?

a. Need for exploration of the ipsilateral kidney
b. Patient with previous suprapubic and/or perineal surgery
c. Patient wishing to maintain potency
d. Patient with bilateral large seminal vesicle lesions
e. Patient with metastatic tumor to the seminal vesicle

31. In a patient with a seminal vesicle abscess, the treatment of choice is:
    a. laparoscopic unroofing.
    b. transvesical excision of the seminal vesicle.
    c. aspiration and antibiotic instillation.
    d. endoscopic unroofing by deep transurethral resection.
    e. retropubic approach to unroof the abscess.

Answers

1. **c. Superior vesicle artery.** The superior vesicle artery does not supply the vas deferens, whereas all of the other arteries listed may have a branch to the vas deferens.
2. **d. It results in a lower rate of complications, including hematoma and infection.** This method eliminates the scalpel incision, results in fewer hematomas and infections, and leaves a much smaller wound than conventional methods of accessing the vas deferens for vasectomy.
3. **a. Hematoma.** The no-scalpel technique significantly decreases the rate of hematomas, infections, and pain during the procedure.
4. **a. Less than 1%.** Vasectomy failure rate when both the abdominal and testicular ends of the divided vas deferens are occluded with hemoclips is less than 1%.
5. **c. Fascial interposition of dartos fascia between the divided ends of the vas deferens. Interposition of dartos fascia between the divided ends of the vas deferens is a technique for occlusion that has been reported to reduce the recanalization rate to nearly zero.**
6. **b. Not excising a long segment of vas deferens.** The technical aspects when performing vasectomy to make vasectomy reversal

easier in the future include not excising a long segment of vas deferens, dividing the vas deferens approximately 3 cm cephalad to the cauda of the epididymis in the straight portion of the vas deferens, and transecting the vas deferens, followed by low-voltage cautery occlusion and then by fascial interposition.

7. **e. A 10% incidence of chronic scrotal pain. Vasectomy does not have an established association with prostate cancer, dementia, cardiovascular disease, or atherosclerosis, although it has been associated with a 10% incidence of chronic scrotal pain.**
8. **d. 60% to 80%. Vasectomy disrupts the blood–testis barrier, resulting in detectable levels of serum antisperm antibodies in 60% to 80% of men.**
9. **b. Motile sperm found in semen analysis 3 months after vasectomy.** If any motile sperm are found in the ejaculate 3 months after vasectomy, consideration should be given to repeating the procedure.
10. **c. The epididymis.** The brunt of pressure-induced damage after vasectomy falls on the epididymis and efferent ductules.
11. **d. Denervation of the cord may offer relief in selected cases.** Microsurgical total denervation of the spermatic cord is a procedure used with reported success in several small series.
12. **e. The Jaboulay bottleneck operation is associated with a low recurrence rate. The Jaboulay bottleneck operation, in which the sac edges are sewn together behind the cord, reduces the chance of recurrence caused by reapposition of the edges of the hydrocele sac.**
13. **b. Adenomatoid tumor.** Most nontransilluminable solid epididymal masses are benign adenomatoid tumors.
14. **a. There was a palpable epididymal abnormality.** A retrospective review of men who underwent epididymectomy for chronic epididymitis showed that outcomes were best when the patient had a palpable epididymal abnormality on physical examination. Men in this study without a palpable abnormality, but with sonographic changes, had slightly worse outcomes, and those with neither a palpable abnormality nor a demonstrable ultrasonographic abnormality did not improve with epididymectomy.

15. **b. A dartos pouch orchidopexy is the treatment of choice.** Creation of a dartos pouch will keep the testis well down into the scrotum and permanently prevent retraction.
16. **a. Excision of the hydrocele sac.** Excising the hydrocele is recommended for long-standing, thick-walled, loculated hydroceles.
17. **d. Are steadily being prescribed more frequently empirically.** Despite evidence that up to 75% of patients with epididymitis/orchitis do not have an identifiable bacterial urinary tract infection concomitantly with their clinical epididymitis, antibiotics are routinely given. Empirical antibiotic administration in the absence of positive urine cultures has been steadily increasing, from 75% to 95% between the years of 1965 and 2005 and is not indicated.
18. **c. 50% of the time.** When a clinically palpable varicocele is encountered in a patient with orchialgia, varicocelectomy will resolve the pain 50% of the time.
19. **c. Distal mesonephric duct.** The seminal vesicle develops as a dorsolateral bulbous swelling of the distal mesonephric duct at approximately 12 fetal weeks.
20. **c. 60% to 80%.** The secretions from the seminal vesicle contribute 60% to 80% of the ejaculate volume.
21. **b. Vesiculodeferential artery.** The blood supply to the seminal vesicle is from the vesiculodeferential artery, a branch of the superior vesical artery.
22. **a. Inflammation of the seminal vesicles.** Seminal vesiculitis shows decreased signal intensity on the T1-weighted image, whereas the T2-weighted image intensity is higher than that of both fat and the normal seminal vesicle.
23. **d. There was an embryologic insult to the mesonephric duct earlier than at 7 weeks' gestation.** Unilateral agenesis of the seminal vesicles has an incidence of 0.6% to 1% and may be associated with unilateral absence of the vas deferens, as well as ipsilateral renal anomalies.
24. **a. Cystic fibrosis. Of men with bilateral absence of the vas deferens or seminal vesicles, 70% to 80% carry the genetic**

mutation associated with cystic fibrosis. Conversely, 80% to 95% of men with cystic fibrosis have bilateral absence of the vas deferens or seminal vesicles.

25. **b. Obstruction of the ejaculatory duct.** Cysts of the seminal vesicles may be either congenital or acquired and are thought to be due to obstruction of the ejaculatory duct.
26. **d. Metastatic tumors. Very few primary tumors of the seminal vesicles have been reported. It is more common for carcinoma of the bladder, prostate, or rectum, or lymphoma to secondarily involve the seminal vesicles.**
27. **b. Transrectal ultrasonography.** Transrectal ultrasonography is the preferred initial test for seminal vesicle abnormality because of its low invasiveness, ease of performance, and ability to perform concomitant transrectal biopsies.
28. **a. Biopsy of the lesion.** Transrectal ultrasonography and biopsy of the seminal vesicle mass is accurate and easily accomplished.
29. **c. The laparoscopic route so that the ipsilateral kidney can be dealt with concomitantly and recovery may be shorter.** Although data are limited for laparoscopic excision of benign seminal vesicle disease alone, this approach appears to afford superb visualization with minimal postoperative morbidity and shorter hospitalization, compared with the open surgical alternatives.
30. **b. Patient with previous suprapubic and/or perineal surgery.** In individuals for whom the perineal or supine position may be difficult to maintain, or for those who have had multiple suprapubic or perineal surgeries, the transcoccygeal approach may be useful.
31. **d. Endoscopic unroofing by deep transurethral resection.** If the abscess is in the portion of the seminal vesicle adjacent to the prostate, a deep transurethral resection into the prostatic substance, just distal to the bladder neck at the 5-o'clock or 7-o'clock position, may be effective in relieving the problem. However, a CT-guided aspiration and drain placement is becoming the preferred least-traumatic option.

Chapter Review

1. Because scrotal cases are considered clean rather than sterile, prophylactic antibiotics are recommended preoperatively. Hair removal should occur immediately before the procedure.
2. Fournier gangrene is a necrotizing fasciitis that involves the skin and subcutaneous tissue and is confined by the dartos fascia on the penis, Colles fascia in the perineum, and Scarpa fascia in the abdomen. Proper resuscitation requires broad-spectrum antibiotics, including a third-generation cephalosporin, an aminoglycoside, and metronidazole. These patients require fluid resuscitation and, when hemodynamically stable, debridement. Daily debridement in the operating room until all nonviable tissue is removed should be subsequently performed.
3. Ninety-seven percent of patients undergoing open-ended vasectomy develop sperm granulomas.
4. Division of the vas deferens during vasectomy should occur at least 3 cm from the epididymitis. There is no vasectomy technique that is 100% effective; more than 80% of the patients achieve azoospermia at 3 months following vasectomy.
5. When the testis is removed for orchialgia, pain relief is better achieved if the orchiectomy is performed through an inguinal incision rather than a transscrotal incision.
6. There is no level 1 evidence that orchiectomy is effective for the treatment of chronic orchialgia.
7. Any surgical manipulation of the epididymis results in azoospermia on that side.
8. Leaving a scrotal drain after scrotal procedures does not lessen the complication rate or the development of postoperative hematomas.
9. When repairing large hydroceles, the epididymis and spermatic vessels may be splayed by the hydrocele, and care must be taken to identify them to avoid injury.
10. Microsurgical denervation of the spermatic cord has been used for the treatment of orchialgia with reported success rates as high as

two-thirds achieving pain relief. It should only be considered if a cord block is successful.

11. Seminal vesicle cysts are associated with ipsilateral renal agenesis or dysplasia in two-thirds of patients and have been associated with polycystic kidney disease.
12. *Mycobacterium tuberculosis* and *Schistosoma haematobium* may infect the seminal vesicles.
13. Vasectomy has not been established to be associated with prostate cancer, dementia, cardiovascular disease, or atherosclerosis, although it has been associated with a 10% incidence of chronic scrotal pain.
14. Vasectomy disrupts the blood-testis barrier, resulting in detectable levels of serum antisperm antibodies in 60% to 80% of men.
15. The Jaboulay operation, in which the sac edges are sewn together behind the cord, reduces the chance of recurrence caused by reapposition of the edges of the hydrocele sac.
16. Of men with bilateral absence of the vas deferens or seminal vesicles, 70% to 80% are carriers of the genetic mutation associated with cystic fibrosis. Conversely, 80% to 95% of men with cystic fibrosis have bilateral absence of the vas deferens or seminal vesicles.
17. Very few primary tumors of the seminal vesicles have been reported. It is more common for carcinoma of the bladder, prostate, or rectum or lymphoma to secondarily involve the seminal vesicles.